

Special Topics in DevOps Lab-3 Tokenization

September 17, 2026

1 Introduction

In this lab, we will utilize software dataset prepared in the last lab.

2 Exercises

1. You are required to write three tokenizers-character, subword, and word level. Analyze the tokenizers around vocab size, average sequence length, embedding matrix size.
2. Select subword tokenizer, initialize a random embedding matrix. Now, find top 50 most frequent tokens in the dataset, top 10 most similar embeddings, top 10 least similar embeddings.
3. Select any 5 tokens from top 50 (make sure at least two tokens are not similar to the student next to you). List the most similar tokens corresponding to it (as per your understanding).
4. Check the similarity between these pairs of tokens.
5. Manually outline a naive algorithm to improve the similarity of related tokens. Then use this algo as prompt to generate code.
6. Again repeat the experiment. write your observations in the README.md.

3 Report and Submission

You would be required to commit and push source code and data on the github repository.

1. Your first few prompts (for each experiment) will be required to be uploaded in output (screenshots)
2. Your first few commits will be evaluated for message quality and significant milestone in the process

3. Upload all other relevant screenshots in output folder
4. README.md
 - Output three statistics-one for each dataset-in the output folder
 - Analyze the dataset and list the problems where these datasets can be used
5. Final submission deadline Sept-23